

Scaling AI Education in Resource-Constrained Environments

A Pilot-Based, Faculty-Centered Model for Institutional Adoption

Suleyman Uludag



College of Innovation & Technology, University of Michigan–Flint

NAIRR Pilot Classroom Expansion Conference

June 18–19, 2026 · Orlando, Florida



Outline

- 1 The Hook: Something Has Already Changed
- 2 Who We Are: Context and Credibility
- 3 The Core Problem
- 4 Our Answer: The AWR² Framework
- 5 The Implementation Model
- 6 Governance: Light but Real
- 7 NAIRR Connection
- 8 Honest Assessment and What's Next

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The Disruption Isn't Coming — It Already Arrived

Industry Signals

~30%

of code now
written by AI

Microsoft CEO

Education Signals

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Student survey, Dec 2024

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- AI can now generate the **artifacts we use to measure learning**: code, reports, essays

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The Punchline

We were updating our syllabi while the workforce restructured around us.

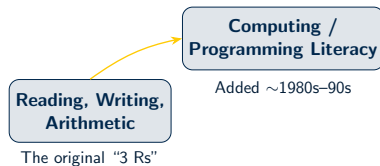
The question is no longer whether to respond — it is how to respond thoughtfully.

AI Fluency Isn't New — It's the Next Rung

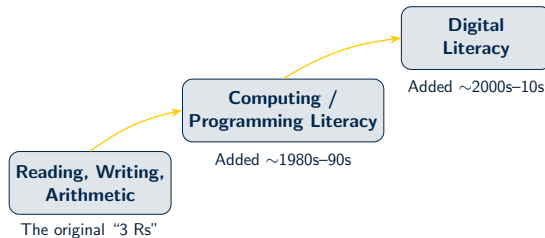
**Reading, Writing,
Arithmetic**

The original "3 Rs"

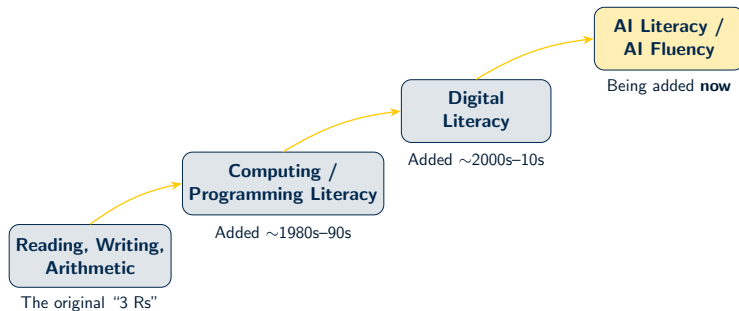
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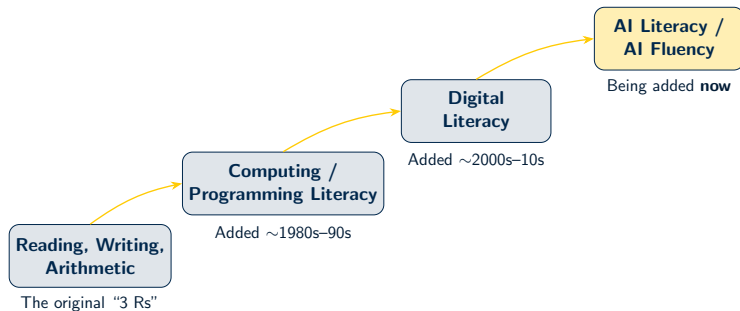
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The Pattern Repeats

Calculators, personal computers, the internet, smartphones — each expansion of “literacy” felt urgent and contested when it arrived.

AWR² treats AI fluency as the next expected rung — not an emergency.

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Who We Are — And Why That Matters Here

UM–Flint College of Innovation & Technology

- Regional public university

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CRADLE Research Group

Cybersecurity Research in **AI**, **Defense**,
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- AI/ML for cybersecurity

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Sound Familiar?

Our main question and challenge: What can we do in a resource-constrained institution trying to do more with less

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Three Problems Landing at Once

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Code, essays, lab reports — produced in seconds by free tools. Traditional assessment is now *leaky*.

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Some ban AI. Some embrace it. Most are somewhere in between. Result: *fragmentation* across courses, programs, and institutions.

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Without coordination, AI adoption risks becoming: uneven, inequitable, and unrepeatable.

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The Implication

We are moving from **individual improvisation** to a need for **coordinated institutional response** — under constraints of limited faculty time, uneven expertise, and evolving infrastructure.

The Resistance Is Real — And Not Unreasoned



1120 Ruthven Building
1109 Geddes Avenue
Ann Arbor, MI 48109

The University of Michigan
University Senate
Motion #042726-2

A Call for GenAI-Free Zones

Whereas in November 2025 the Faculty Senate [overwhelmingly endorsed a resolution](#) “demanding a comprehensive policy addressing the role of Generative Artificial Intelligence at the University of Michigan”;

Whereas there is no mechanism by which we as teachers can know whether a student has used GenAI in the composition of an essay, a problem set, or any other piece of work, making it impossible for us to accurately assess anything that a student does outside the classroom context;

Whereas GenAI thereby short-circuits the work of education, making it too easy for even the most well-intentioned students to take shortcuts:

... four additional clauses omitted ...

Therefore, be it resolved that the Faculty Senate urges the Provost and the U-M administration to invest in the creation of infrastructure designed to make it easier for faculty to manage if, when, and how GenAI platforms can be accessed by students based on the learning goals set by

facultysenate.umich.edu/university-senate-vote-results-april-27-2026

University of Michigan Senate

Motion #042726-2 — “A Call for GenAI-Free Zones”

Passed April 30, 2026

Yes: 1,711 (82.8%) · No: 356 (17.2%) · Abstain: 269

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Why This Matters for AWR²

Exactly why Faculty Development is **opt-in, not mandatory** — structure without rigidity means room for colleagues who opt out of GenAI.

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AWR²: AI-Ready, Work-Ready

A Coordinated Institutional Posture — Not Just a Curriculum

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**AI
Literacy**

What AI can
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Grade process,
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Opt-in, supported,
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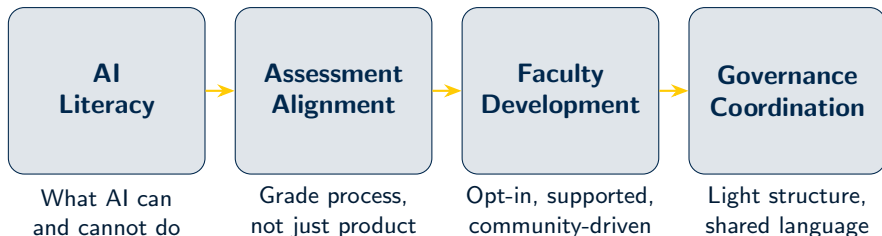
Opt-in, supported,
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Governance Coordination

Light structure,
shared language

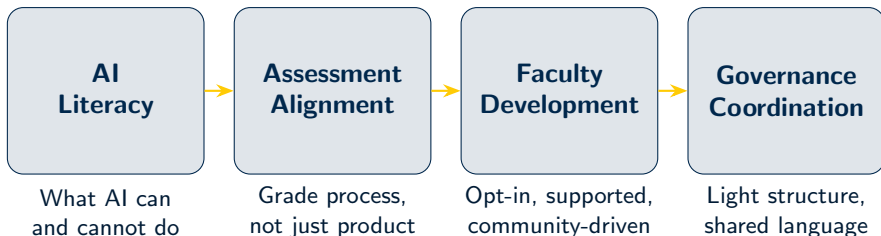
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Origin

Emerged from the **CIT AI Teaching Circle** (10 faculty, Fall 2025–Spring 2026) and an institutional proposal to the UM Strategic Initiative Fund. Grounded in faculty-led inquiry, not top-down mandate.

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From Strategy to Action: Pilot → Learn → Refine → Scale

Strategic Direction (AWR²)

- (1) AI literacy (2) Assessment alignment (3)
Faculty development (4) Governance coordination

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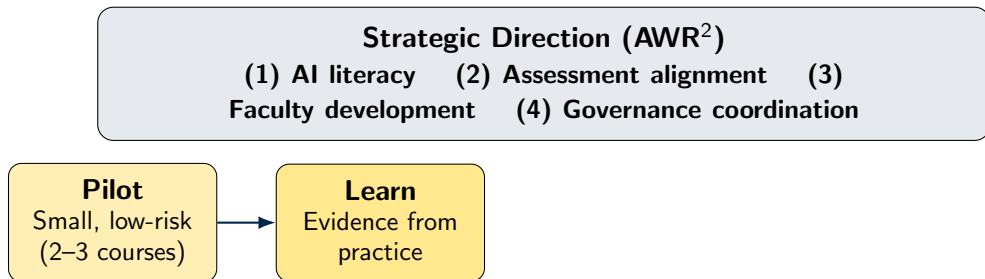
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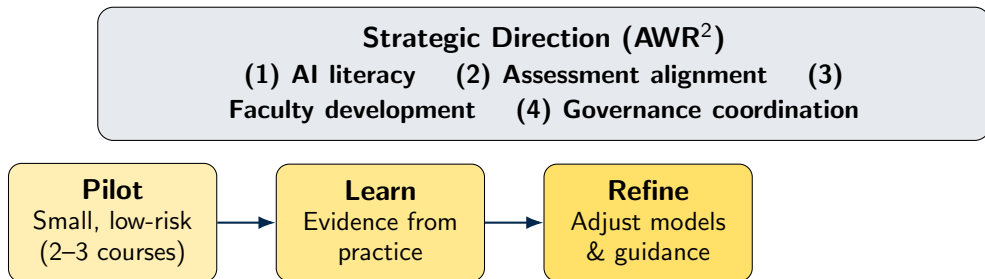
Pilot

Small, low-risk
(2–3 courses)

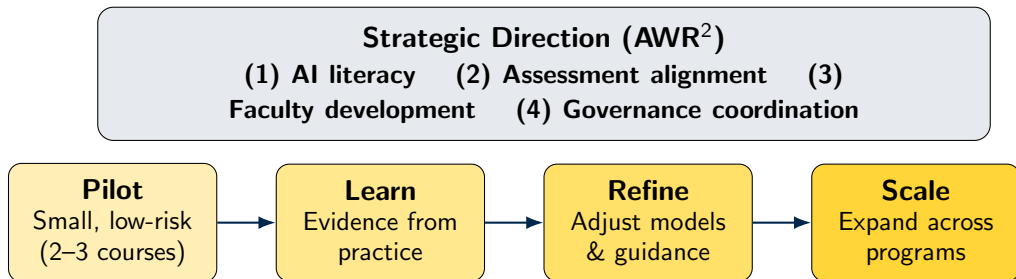
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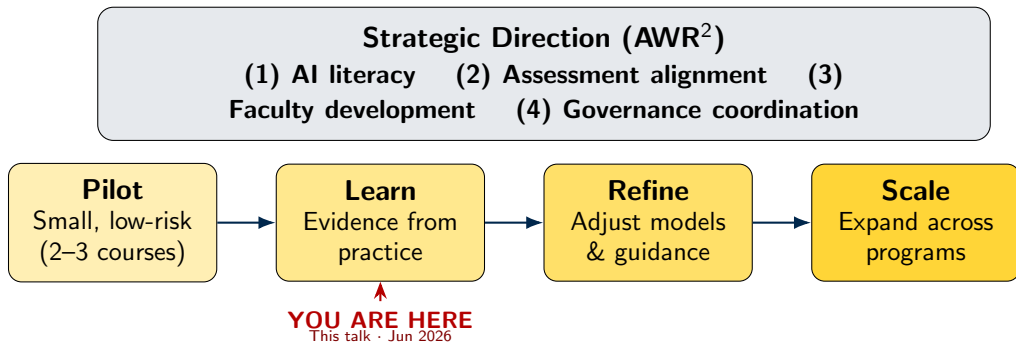
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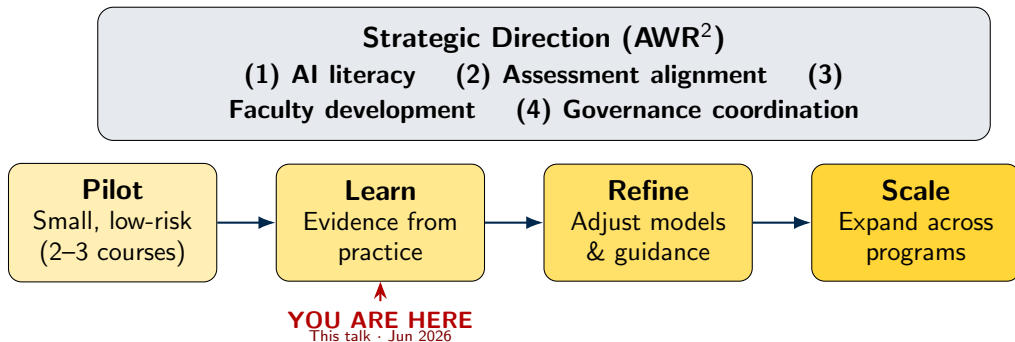
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The Key Insight

Think of it as **control theory applied to curriculum change**: pilot activities generate evidence, feedback informs refinement, iterative adjustments guide scaling decisions — analogous to reward and penalty signals in adaptive systems.

Start small. Learn quickly. Scale deliberately.

What “Pilot” Actually Looks Like: CS1 Fall 2026

The Courses

Course	Lang	Priority
CSC 122	Python	High
CSC 175	C/C++	High
CSC 176	Java	High
CSC 275	C/C++	Medium
CSC 276	Java	Medium

<5 volunteer faculty · one semester · opt-in only

Scope: undergrad CS1/CS2 only; master's-level is a planned Refine/Scale target

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The Pedagogical Spine

Adapted from Porter & Zingaro (ITiCSE 2024); Biswas & Ahmed (2026):

- ① **Plan** — before AI is invoked
- ② **Approve** — instructor/peer checklist
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- ⑤ **Reflect** — accept? modify? reject?

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The Core Assessment Shift

We grade **the process** — plan quality, verification evidence, reflection honesty — not the artifact alone.
“It worked when I tried it” is no longer acceptable evidence of learning.

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“It worked when I tried it” is no longer acceptable evidence of learning.

Evidence: in a UVA pilot, students with full AI access on a reading task barely talked — they assumed they had the answer; the no-AI group was most engaged. (The Chronicle, Jun 2026)

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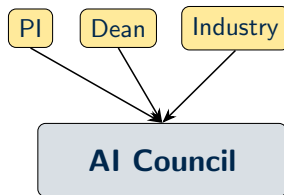
Governance: Coordinating Without Controlling



AI Council

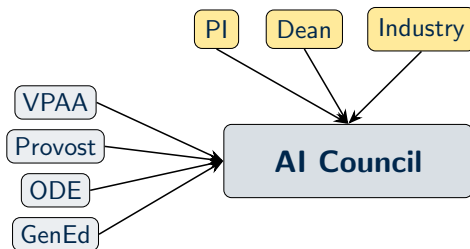
The AI Council is the coordination center — not a gatekeeping body.

Governance: Coordinating Without Controlling



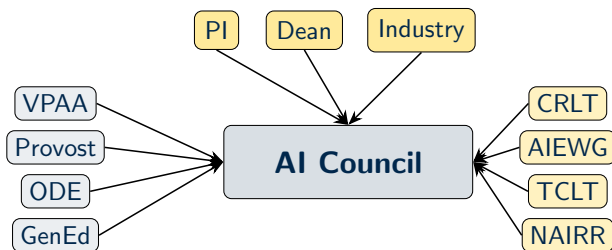
Leadership gives the initiative authority without making it top-down.

Governance: Coordinating Without Controlling



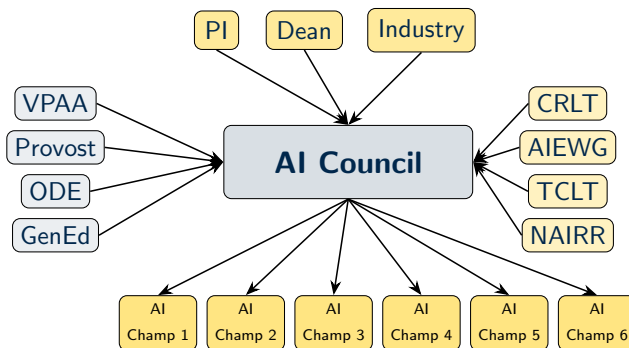
Academic alignment without new approval layers.

Governance: Coordinating Without Controlling



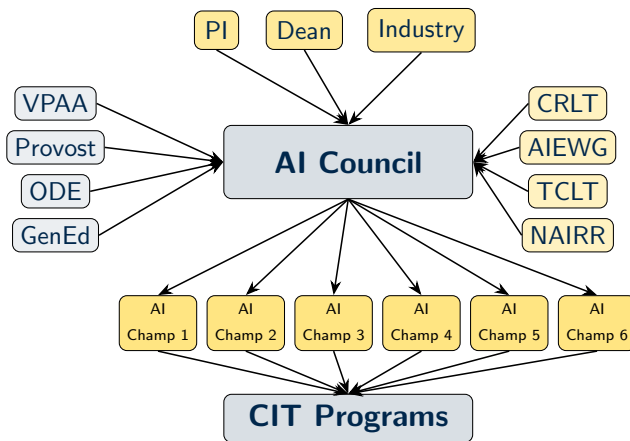
Teaching, learning, and NAIRR partners strengthen capacity.

Governance: Coordinating Without Controlling



AI Champions translate strategy into disciplinary practice.

Governance: Coordinating Without Controlling



Structure without rigidity. Experimentation without chaos.

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NAIRR: Reducing the Infrastructure Barrier

Our NAIRR Experience

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- Students run real ML/security experiments without personal hardware costs
- **NAIRR democratizes access** — the single biggest infrastructure barrier for institutions like ours

NAIRR: Reducing the Infrastructure Barrier

Our NAIRR Experience

- NSF EAGER Award #2515085: *AI/ML for Cybersecurity Education via NAIRR*
- Used NAIRR Pilot for:
 - GPU compute for student labs
 - Jupyter environments
 - Dataset access for research
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When NAIRR Isn't Available Yet

For our CS1 pilot, we are also deploying a **locally hosted GPU workstation** as a bridge:

- Dell Precision 7875, RTX 6000 Ada (48GB)
- OpenWebUI front-end for open-weight models
- Every student: same model, same terms, no subscription cost
- Policy compliant: student data stays on UM infrastructure

Equity through centralized access — not individual subscriptions

Roadmap: as the pilot scales, we plan to request a NAIRR classroom allocation so CS1 runs on the same infrastructure already serving our research students

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Outline

- 1 The Hook: Something Has Already Changed
- 2 Who We Are: Context and Credibility
- 3 The Core Problem
- 4 Our Answer: The AWR² Framework
- 5 The Implementation Model
- 6 Governance: Light but Real
- 7 NAIRR Connection
- 8 Honest Assessment and What's Next

Where We Are: Honest Signals, Not Declarations of Victory

What We Have

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- Large-scale outcomes data
- Full institutional adoption
- A perfect governance model
- Evidence the model works **outside CS/computing**
- Pilot coverage at the **master's level**
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“No playbook out there . . . nobody has figured it out yet.”
— Leo Lo, Dean of Libraries, UVA (Chronicle, Jun 2026)

Three Things You Can Do Monday Morning

1. **Start with 2–3 volunteer faculty — not a mandate, not a committee**

Find the people already experimenting. Give them a name, a structure, and a small amount of support. The Teaching Circle model works.

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The Underlying Principle

Structure without rigidity. Experimentation without chaos.

You do not need a finished plan. You need a direction, a feedback loop, and willing colleagues.

Let's Build Something Together

I'm Looking to Connect With People Who Are:

- Wrestling with **assessment validity** in the age of generative AI

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Find me
after this talk



uludag@umich.edu

Concrete Collaboration Opportunities

- Share pilot materials (rubrics, templates, syllabi)
- Co-author FIE / SIGCSE / ASEE papers
- Multi-institution NSF proposals
- Cross-institutional NAIRR classroom projects

Thank You

Questions, Comments, Pushback?

Suleyman Uludag

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**Classroom
Expansion
Conferences**
- Research Emerging -
Large

Backup: Three Roads — How Peers Are Responding

Purdue

Top-down mandate. All 40,000+ undergrads, 200+ majors, effective Fall 2026.

“A press release, not a plan” — logistically impossible at this scale and speed.

Inside Higher Ed, Dec 2025

Ohio State

Large-scale, faculty-grounded. 100 AI faculty hired; six universitywide learning outcomes.

Grown from “Buck AI,” a cross-campus working group active since 2023.

The Chronicle, Jun 2026

UVA

Volunteer, pilot-based. “AI Literacy and Action Lab”; 5-domain framework (technical, ethical, critical thinking, practical, societal).

“No playbook out there . . . nobody has figured it out yet.”

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Where AWR² Sits

Closest to UVA's pilot-first posture — but adds an explicit governance layer (AI Council, AI Champions) and a disciplined Pilot → Learn → Refine → Scale feedback loop that UVA's model doesn't yet formalize.